Akshay Patil

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EDUCATION

Georgia Institute of Technology – Atlanta, GA **August 2026 – May 2029**
Bachelor of Science in Aerospace Engineering
Current coursework: AE 1601 Aerospace Engineering, COE 2001 Statics, MSE 2001 Materials, MATH 2551 Multivariable Calculus, ME 1670 Graphics, CS 1371 Computing

School of Science and Engineering at Townview – Dallas, TX **August 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | NASA High School Aerospace Scholar

EXPERIENCE

Salvo Dynamics | Design Engineer **August 2026 – Present**
- Salvo builds small interceptor aircraft meant to fly out and bring down hostile drones, and is separately building an RC jet to break the speed of sound
- Turning optimized aerobody models into load-bearing structures in CAD, and laying out the avionics that go inside them
- Designing 3D printed airframes for testing, plus the assembly jigs the production aircraft are built on

Hardpoint Robotics | Co-Founder **June 2026 – Present**
- Hardpoint builds NDAA-compliant development UAVs, aircraft for customers who are not allowed to fly foreign-sourced hardware
- I own the airframe: CAD, 3D printed CF-Nylon parts, and carbon fiber structures made with outside vendors
- Designing around sourcing as a hard constraint, since every part has to be compliant and purchasable, so vendor lead times shape the design as much as the loads do

SWYFT Robotics | Engineering Design Intern **May 2025 – May 2026**
- SWYFT sells competition drivetrain and structural parts to FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed sprockets, gears, and pulleys that hold up under competition loads at a price teams can afford
- Designed a modular robot cart for FRC teams with a detachable programming station, drawn for extrusion and fiber laser cutting

RESEARCH AND CLUBS

Hypersonics VIP, Georgia Tech | Data and Communications Team **August 2026 – Present**
- The team is standing up a sonic fatigue test facility, where acoustic loading is used to find how structures fail under sustained high-intensity sound
- Building the sensor package that measures pressure, temperature, and airflow inside the facility
- Developing parabolic mirrors to generate a uniform sound field across the test section

Georgia Tech Experimental Rocketry | Externals Team **August 2026 – Present**
- The externals team builds the rocket's prepreg carbon fiber structure in house, which I joined to learn composite layup
- Designing a fin mounting jig that locates fins on an already wrapped tube whose exact dimensions are no longer known, and holds fin cant to within a tenth of a degree
- Working out how to measure each fin angle to that tolerance, with a mini carbon fiber fin can going to a third party for dimensioning to validate the jig

Yellow Jacket Space Program | Turbomachinery, Engine Development **August 2026 – Present**
- The program is going for the collegiate liquid propellant altitude record and the Karman line at 100 km
- Designing electric pumps and turbopumps so the vehicle can drop the large, heavy pressurant tanks a pressure-fed engine needs
- Designing and manufacturing impellers, pump housings, and the surrounding hardware

IVALab, Georgia Tech | Undergraduate Researcher **September 2026 – Present**
- The lab works on navigation for robots that have to move through crowds of people
- Developing navigation algorithms that mix learned policies with classical planners
- Benchmarking them against each other in the Arena simulator

FIRST Tech Challenge Team 7172 | Captain and Lead Designer **September 2021 – May 2026**
- Lead designer on four robots; 2026 World Championship Finalist Alliance, 19 event wins and 63 awards over five seasons
- Designed a 6-DoF arm robot that expands from an 18 in cube to 13 ft of reach: three-pivot arms on worm-gear turrets, torsion springs offloading the pivot motors, and a drivetrain that splits in half on linear slides, scoring on roughly 4 s cycles

SKILLS

CAD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor
Manufacturing: 3D printing (CF-Nylon), fiber laser cutting, extrusion, carbon fiber vendor work
Programming: Python (NumPy, Matplotlib, Pandas, TensorFlow), MATLAB, JavaScript, SQL, Git
Hardware: NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
Certifications: FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)